

Main content (from notes):

Role of Water in the Chemical Properties of Acids

- The chemical properties of acids are dependent on the presence of hydrogen ions (H^+) produced by the dissociation of acid molecules in water.
- It is the hydrogen ions that gives acids its ability to react with reactive metals, bases and carbonates.
- This explains why a solution of hydrogen chloride dissolved in ethanol does not show acidic properties. Hydrogen chloride dissolves in chloromethane but remains undissociated.
- acids are neutral compounds that ionize when they dissolve in water to give H^+ ions and a corresponding negative ion (1)

Reaction 1: acid + reactive metal $\rightarrow$ salt + hydrogen gas

Eg 1: Dilute hydrochloric acid with magnesium

Chemical equation: $2HCl(aq) + Mg(s) \rightarrow MgCl_2(aq) + H_2(g)$

Ionic equation: $2H^+(aq) + Mg(s) \rightarrow Mg^{2+}(aq) + H_2(g)$

Eg 2: Dilute sulfuric acid with zinc

Chemical equation: $H_2SO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + H_2(g)$

Ionic equation: $2H^+(aq) + Zn(s) \rightarrow Zn^{2+}(aq) + H_2(g)$

*Note: Some (unreactive) metals will not react with dilute hydrochloric and sulfuric acids to form hydrogen gas. e.g. copper, silver, gold, platinum and mercury
(This section will be dealt with in detail in "Reactivity Series", under the topic, "Metals")

*Note: Nitric acid reacts with some metals (e.g. copper) to evolve a brown gas, nitrogen dioxide (NO_2) instead of hydrogen due to redox reaction.

explanation #1

- as shown by the ionic equations, the H^+ ion is the enabler, allowing acids to show their acidic properties (ex: in rxn1, even after the reaction, Cl^- is still unchanged, while the H^+ ion has reacted with Mg to become H_2)
- without water, the acid is just a neutral compound and therefore cannot engage in chemical reactions

Reaction 2: acid + carbonate → salt + carbon dioxide gas + water

Eg 1: Dilute sulfuric acid with copper(II) carbonate

Chemical equation: $\text{CuCO}_3 (\text{s}) + \text{H}_2\text{SO}_4 (\text{aq}) \rightarrow \text{CuSO}_4 (\text{aq}) + \text{CO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l})$

Ionic equation: $\text{CuCO}_3 (\text{s}) + 2\text{H}^+ (\text{aq}) \rightarrow \text{CO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l}) + \text{Cu}^{2+} (\text{aq})$

Eg 2: Dilute hydrochloric acid with aqueous sodium carbonate

Chemical equation: $\text{Na}_2\text{CO}_3 (\text{aq}) + 2\text{HCl} (\text{aq}) \rightarrow 2\text{NaCl} (\text{aq}) + \text{CO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l})$

Ionic equation: $\text{CO}_3^{2-} (\text{aq}) + 2\text{H}^+ (\text{aq}) \rightarrow \text{CO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l})$

Eg 3: Dilute nitric acid with calcium carbonate

Chemical equation: $\text{CaCO}_3 (\text{s}) + 2\text{HNO}_3 (\text{aq}) \rightarrow \text{Ca}(\text{NO}_3)_2 (\text{aq}) + \text{CO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l})$

Ionic equation: $\text{CaCO}_3 (\text{s}) + 2\text{H}^+ (\text{aq}) \rightarrow \text{CO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l}) + \text{Ca}^{2+} (\text{aq})$

Reaction 3: acid + base → salt + water

Recap: some examples of bases are: CuO, NaOH, KOH, etc

Eg 1: Dilute hydrochloric acid with aqueous sodium hydroxide

Chemical equation: $\text{HCl} (\text{aq}) + \text{NaOH} (\text{aq}) \rightarrow \text{NaCl} (\text{aq}) + \text{H}_2\text{O} (\text{l})$

Ionic equation: $\text{H}^+ (\text{aq}) + \text{OH}^- (\text{aq}) \rightarrow \text{H}_2\text{O} (\text{l})$

Eg 2: Dilute sulfuric acid with copper(II) oxide

Chemical equation: $\text{H}_2\text{SO}_4 (\text{aq}) + \text{CuO} (\text{s}) \rightarrow \text{CuSO}_4 (\text{aq}) + \text{H}_2\text{O} (\text{l})$

Ionic equation: $2\text{H}^+ (\text{aq}) + \text{CuO} (\text{s}) \rightarrow \text{Cu}^{2+} (\text{aq}) + \text{H}_2\text{O} (\text{l})$

Eg 3: Dilute nitric acid with zinc oxide

Chemical equation: $2\text{HNO}_3 (\text{aq}) + \text{ZnO} (\text{s}) \rightarrow \text{Zn}(\text{NO}_3)_2 (\text{aq}) + \text{H}_2\text{O} (\text{l})$

Ionic equation: $2\text{H}^+ (\text{aq}) + \text{ZnO} (\text{s}) \rightarrow \text{Zn}^{2+} (\text{aq}) + \text{H}_2\text{O} (\text{l})$

Application question: How do you confirm whether an unknown solution contains an acid?

Use litmus paper (blue to red)

Use Universal Indicator (green to red/orange/yellow)

Use pH probe (pH less than 7)

Add reactive metal (hydrogen gas produced)

Add carbonate (carbon dioxide produced)

sources

1. <https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch11/acidbase.php>

slides

1. small introduction of the water (basic outline of role of water and the general idea of dissociation)
2. reaction #1
3. examples for reaction #1
4. reaction #2
5. examples for rxn #2
6. rxn #3
7. rxn #3 examples
8. appli qns
9. qna?
10. thank u